

Lesson 2: Introduction to Simulations

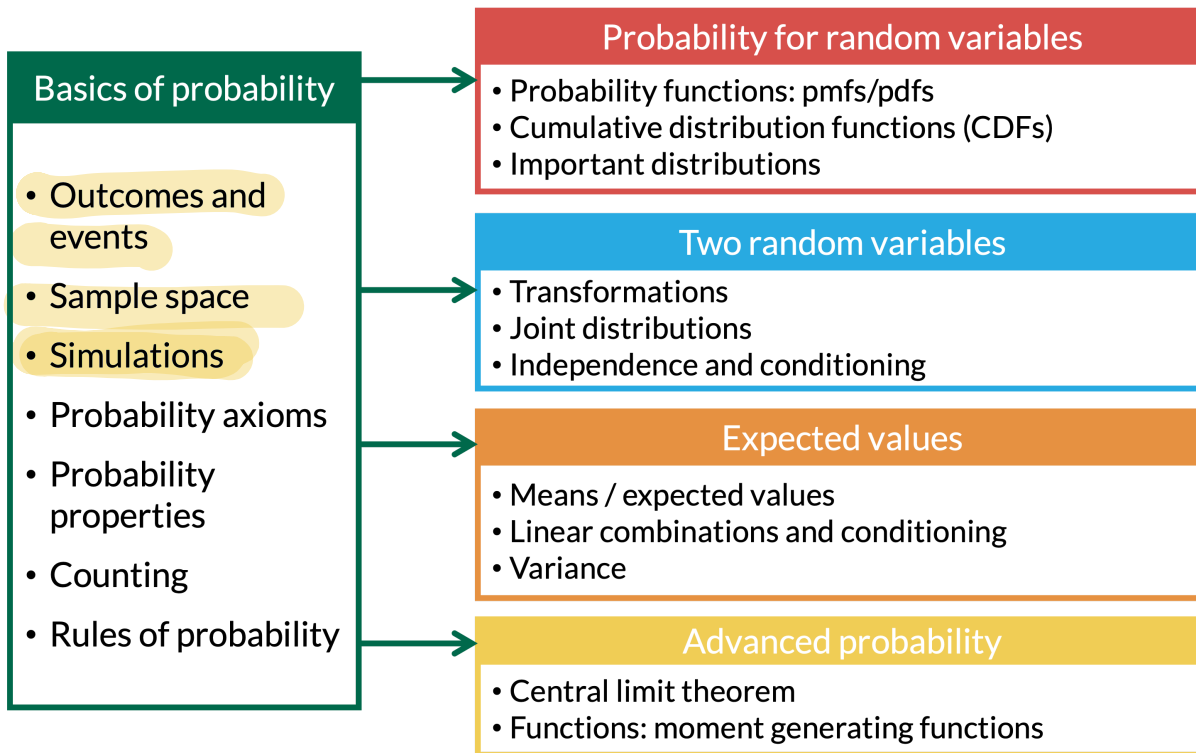
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2026-09-30

Learning Objectives

1. Describe random variables and distinguish between discrete and continuous types
2. Explain the role of simulation in approximating probabilities and distributions
3. Use R to run simulations of discrete random variables
4. Apply and interpret the four-step simulation process

Where are we?



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Recall: Outcomes, events, sample spaces

Definition: Outcome

The possible results in a random phenomenon

→ 1 coin flip

→ 2 coin flips

→ 3 dice rolls

Definition: Sample Space

The **sample space** S is the set of *all* outcomes

Definition: Event

probability related to events

An **event** is a *collection of some outcomes*. An event can include multiple outcomes or no outcomes (a subset of the sample space).

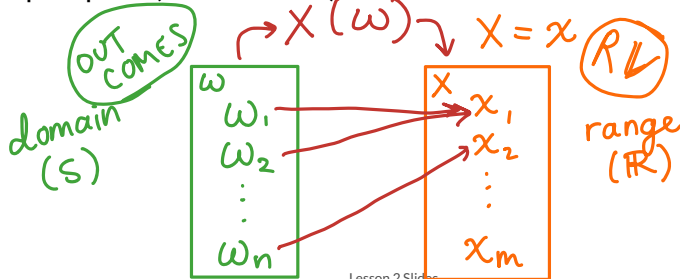
When thinking about events, think about outcomes that you might be asking the probability of. For example, what is the probability that you get a heads or a tails in one flip? (Answer: 1)

We need to understand random variables

Definition: Random Variable

For a given sample space S , a **random variable** (r.v.) is a **function** whose domain is S and whose range is the set of real numbers $\mathbb{R}$. **A random variable assigns a real number to each outcome in the sample space.**

- A random variable's value is completely determined by the outcome ω , where $\omega \in S$
 - What is *random* is the outcome ω
- A **random variable** is a function from the sample space (with outcomes ω) to the set of real numbers
 - We typically write $X(\omega)$ (or X for short), where X is our random variable
- Thus, we can take our sample space (all outcomes) and make functional transformations to it



2 ω 's can
point to same
 x

1 ω cannot
point to more
than 1 x

The cool (and tricky) thing about random variables

Do you remember our coin example from Lesson 1? We tossed one or two coins.

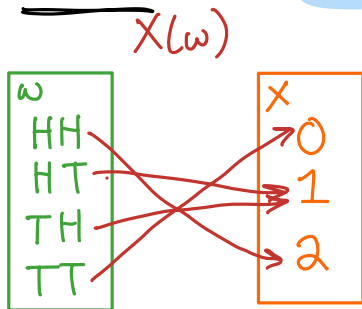
- For each coin, the **sample space** is heads and tails ($S = \{H, T\}$)
- If we want the **sample space** for both coins in order, then we have combinations (S

$$\rightarrow = \{(\underset{1}{H}, \underset{2}{H}), (\underset{1}{H}, \underset{2}{T}), (\underset{1}{T}, \underset{2}{H}), (\underset{1}{T}, \underset{2}{T})\}$$

We make the **random variable** a *function* of the **sample space**.

✗ For one coin toss, we can say random variable X is 1 if we toss a heads ($\omega = H$) and $X = 0$ if we get a tails

- For the two coins, we can say X is the count of heads, so if $\omega = (H, T)$, then $X = 1$



$$X(\omega) = \begin{cases} 0 & \omega = (T, T) \\ 1 & \omega = \{(H, T), (T, H)\} \\ 2 & \omega = (H, H) \end{cases}$$

Types of random variables

There are two types of random variables:

- **Discrete random variables (RVs)**: the set of possible values is either finite or can be put into a countably infinite list
 - You could *theoretically* list the specific possible outcomes that the variable can take
 - If you sum the rolls of three dice, you must get a whole number. For example, you can't get any number between 3 and 4.
 - **Continuous random variables (RVs)**: take on values from continuous *intervals*, or unions of continuous intervals
 - Variable takes on a range of values, but there are infinitely possible values within the range
 - If you keep track of the time you sleep, you can sleep for 8 hours or 7.9 hours or 7.99 hours or 7.999 hours ...
- **Discrete random variables (RVs)** are a little easier to simulate right now
 - We will only do discrete RVs today

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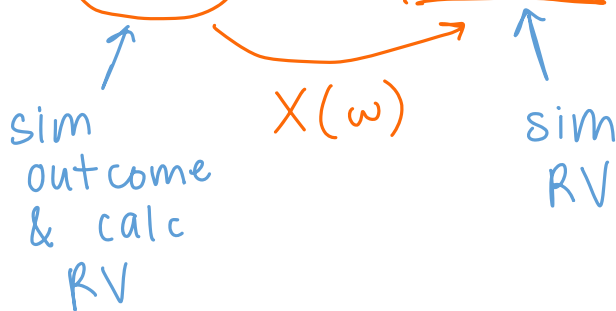
What is a simulation?

A **probability model** for a **random phenomenon** includes a **sample space**, **events**, **random variables**, and a **probability measure**.

Simulation

Simulation involves using a probability model to artificially recreate a **random phenomenon**, many times, usually using a computer.

We simulate outcomes and values of random variables according to the model's assumptions.



The Foundation: Relative Frequencies

- Probabilities can be interpreted as long-run relative frequencies

★ many times : 1000
out of 1000 times,
how often does —
happen?

- By simulating a random phenomenon a large number of times, we can approximate the probability of an event by calculating the relative frequency of its occurrence

★ Basically, out of all the trials we run, how many times did the event happen?

- Simulation is a powerful tool to approximate a few things:
 - Probabilities
 - Distributions of random variables
 - Long-run averages

We saw an example of long-run relative frequency in our coin flip

In Lesson 1, we flipped a coin 100 times and recorded the proportion of heads.

- We tossed 50 heads out of the 100 flips
- Our **long-run relative frequency** was $50/100 = 0.5$, which approximated the probability of getting a head on any one flip

$A = \text{heads}$

Chance Events

Randomness is all around us. Probability theory is the mathematical framework that allows us to analyze chance events in a logically sound manner. The probability of an event is a number indicating how likely that event will occur. This number is always between 0 and 1, where 0 indicates impossibility and 1 indicates certainty.

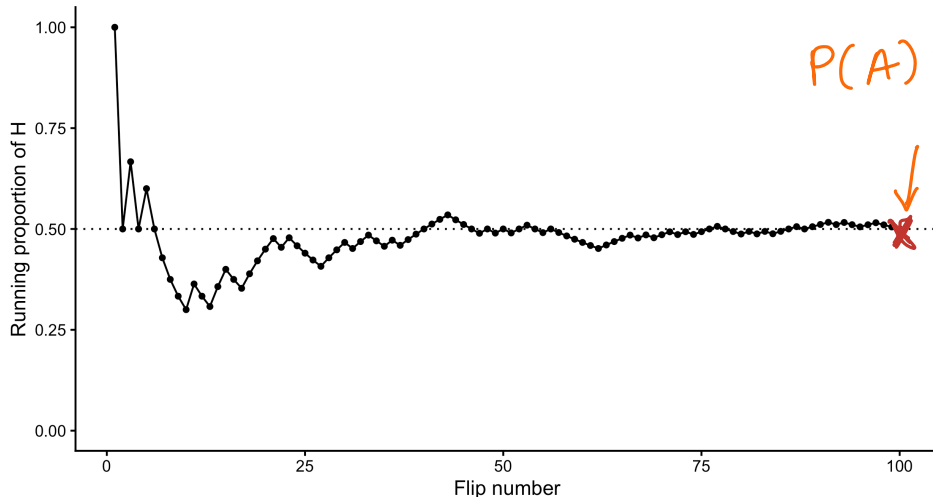
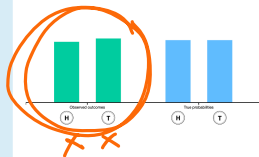
A classic example of a probabilistic experiment is a fair coin toss, in which the two possible outcomes are heads or tails. In this case, the probability of flipping a head or a tail is 1/2. In an actual series of coin tosses, we may get more or less than exactly 50% heads. But as the number of flips increases, the long-run frequency of heads is bound to get closer and closer to 50%.

T

Flip the coin

Flip outcomes

For an unfair or weighted coin, the two outcomes are not equally likely. You can change the weight or distribution of the coin by dragging the true probability bar (on the right in ) up or down. If we assign numbers to the outcomes—say, 1 for heads, 0 for tails—then we have created the mathematical object known as a [random variable](#).



Tactile simulations

- We've already seen coin flips!
- We can also use cards, dice, and other objects to simulate discrete random variables
- Other common method: A **box model** uses a box/hat/bucket of "tickets" with labels to represent possible outcomes
 - Allows us to increase the number of "tickets" with appropriate labels
 - **Coin flip as box model:** A box with two tickets (H and T).
 - **90% free throw shooter:** A box with 10 tickets (9 "make" and 1 "miss").
 - Draws can be **with replacement** (e.g., coin flips) or **without replacement** (e.g., dealing a poker hand).
 - ↳ I take a ticket, record, put it back
 - ↳ don't put back in

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Example to build our simulation skills

Example: Simulating Two Rolls of a Fair Four-Sided Die

We're going to roll two four-sided dice. Let X be the sum of two rolls, and Y be the larger of the two rolls. How would we simulate X and Y separately?

- Note: this example is not asking for a probability!
 - We can simulate a random variable and look at its distribution without calculating any probabilities.
- We will focus on simulating X first

Let's build up some coding tools to do this!

How do we simulate something like a single roll of a die?

- We can also use `R` to sample from the box or spinner
- The `sample()` function is a powerful tool for simulating draws from a box model.
- For example, we can simulate a coin flip
 - What is `x`? → possible outcome
 - What is `size`? → how many draws to take?

★ *sample is best for equally likely outcomes*

```
1 sample(x = c("H", "T"), size = 1)
[1] "T"
```

90% free throw: `c("make", "make", ..., "make", "miss")`
↳ binomial RV (sneak peak)

- Or a roll of a four-sided die

```
1 sample(x = c(1, 2, 3, 4), size = 1)
[1] 1
```

What if we have multiple rolls at once?

- We can set `size` to be larger than 1 to simulate multiple draws at once

```
1 sample(x = c("H", "T"), size = 5, replace = TRUE)
```

```
[1] "T" "T" "T" "H" "T"
```

→ put the H or T back in

- We can simulate our example of the two four-sided dice

```
1 sample(x = c(1, 2, 3, 4), size = 2, replace = TRUE)
```

```
[1] 3 3
```

rolls 1 2

- What happens if we set `replace = FALSE`?

```
1 sample(x = c(1, 2, 3, 4), size = 2, replace = FALSE)
```

```
2 sample(x = c(1, 2, 3, 4), size = 4, replace = FALSE)
```

```
[1] 4 2
```

```
[1] 2 1 3 4
```

we don't replace "tickets"

Can we start to simulate many rolls of two dice?

Example: Simulating Two Rolls of a Fair Four-Sided Die

We're going to roll two four-sided dice. Let X be the sum of two rolls, and Y be the larger of the two rolls. How would we simulate X and Y separately?

- To repeat our pair of rolls many times, we make a **table with one row for every roll**
 - `sim` numbers each simulation (each pair of rolls), and `die` says which die was rolled
 - `expand_grid()` makes every combination of `sim` and `die`, and `n()` is the number of rows

- Let's start with 5 simulations

```
1 reps <- 5
2
3 rolls <- expand_grid(sim = 1:reps, die = 1:2) |>
4 mutate(roll = sample(1:4, n(), replace = TRUE))
```

Handwritten annotations:

- `reps` is circled in red, with an arrow pointing to `1:reps` in line 3.
- `1:reps` is circled in red, with an arrow pointing to `1:4` in line 4.
- `1:4` is circled in red, with an arrow pointing to `n()` in line 4.
- `n()` is circled in red, with an arrow pointing to `10` (handwritten).
- `10` has an arrow pointing to `c(1, 2, 3, 4)` (handwritten).
- `1:4` has an arrow pointing to `c(1, 2, 3, 4, 5)` (handwritten).
- `1:2` is circled in blue, with an arrow pointing to `column: sim & die` (handwritten).
- `new column` (handwritten) has an arrow pointing to the `roll` column in line 4.
- ①, ②, ③ are circled in red on the right side of the code block.

Lesson 2 Slides

What does our table of rolls look like?

1 rolls ①

A tibble: 10 × 3

row#	sim	die	roll	sum
	<int>	<int>	<int>	
1	1	1	1	5
2	1	2	4	
3	2	1	2	3
4	2	2	1	
5	3	1	3	4
6	3	2	1	
7	4	1	3	5
8	4	2	2	
9	5	1	3	5
10	5	2	2	

Handwritten annotations: Blue arrows point from 'row#' to the row numbers. Blue circles highlight the 'sim' column values. Yellow circles highlight the 'die' column values. Yellow arrows point from 'die' to 'roll'. A red vertical line separates the 'roll' column from the 'sum' column. Blue arrows point from the 'roll' column to the 'sum' column. A yellow arrow points from the 'sim' column to the 'sum' column. Red lines are drawn at the bottom.

- Each row is **one roll of one die**
- Each simulation (**sim**) takes up **two rows**: one for **die 1** and one for **die 2**
 - For example, the two rows with **sim** = 1 are the first pair of rolls
- Next, we combine the two rolls in each simulation to get **X**

Sum of
two rolls

Then we calculate X for each simulation

- X is the sum of the two rolls, so we add up the rolls **within each simulation**

- `group_by(sim)` tells R to treat each simulation separately

- `summarise()` collapses each group into one row

sim 1 stays w/ 1
sim 2 stays w/ sim 2

```
1 rolls |>  
2   group_by(sim) |>  
3   summarise(X = sum(roll))
```

A tibble: 5 × 2

	sim	X
	<int>	<int>
1	1	5
2	2	3
3	3	4
4	4	5
5	5	5

→ random variable

- Now we have one row per simulation and one value of X in each row

We need more reps for long-run relative frequencies

- Same code, but now with 10,000 simulations

```
1 reps <- 10000 → ∞
2
3 rolls <- expand_grid(sim = 1:reps, die = 1:2) |>
4   mutate(roll = sample(1:4, n(), replace = TRUE))
5
6 X_sims <- rolls |>
7   group_by(sim) |>
8   summarise(X = sum(roll))
```

- Let's look at the first few simulations

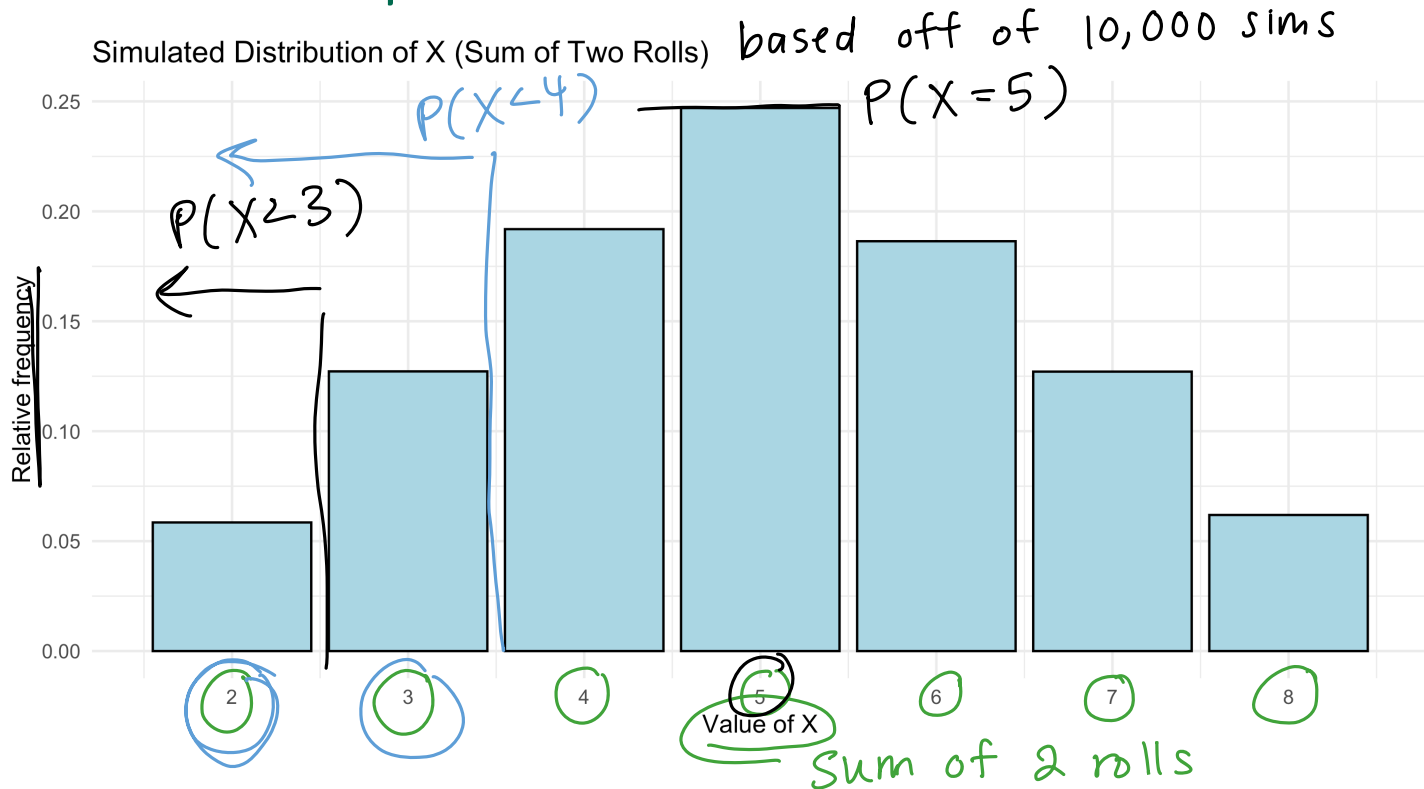
①

```
1 X_sims
# A tibble: 10,000 × 2
   sim      X
   <int> <int>
1     1     5 → T
2     2     4 → F
3     3     7 → F
4     4     2 → F
5     5     7 → F
6     6     5 → T
7     7     8
8     8     5
9     9     3
10    10     5
# i 9,990 more rows
```

②

③

We can look at the plot of random variable X



If we want to calculate something else, we can!

- `summarise()` can calculate several summaries at once:

- Average and standard deviation of X
- Probability that $X = 5$ and probability that $X < 3$

```
1 X_sims |> summarise(  
2   average = mean(X)  
3   std_dev = sd(X)  
4   prob_5 = mean(X == 5)  
5   prob_less_3 = mean(X < 3)  
6 )  
7
```

A tibble: 1 × 4

	average	std_dev	prob_5	prob_less_3
	<dbl>	<dbl>	<dbl>	<dbl>
1	5.00	1.57	0.247	0.0585

Handwritten notes and calculations:

- $P(X=5)$ (circled in green)
- $P(X < 3)$ (circled in orange)
- $\text{sum}(X == 5) / \text{reps}$ (with arrows pointing to `prob_5` and `0.247`)
- $P(X < 3) = 0.0585$ (with an arrow pointing to `prob_less_3`)
- $P(X=5) = 0.247$ (with an arrow pointing to `prob_5`)

Probabilities (**relative frequencies**) are the **proportion of simulations** where the event occurs:

- $X == 5$ is **TRUE** or **FALSE** for each simulation
- R treats **TRUE** as 1 and **FALSE** as 0, so $\text{mean}(X == 5)$ is the number of times the event occurs divided by the total number of simulations (reps)

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4 (S)teps of a Simulation

1. Set up

Define the probability space and related random variables and events, including assumptions.

$c(1,2,3,4)$ equally likely

2. Simulate

Run the simulation to generate outcomes according to the assumptions.

$\text{sample}()$

3. Summarize

Analyze the output using plots and summary statistics like relative frequencies and averages.

~~4~~ 4. Sensitivity analysis

Investigate how results change when assumptions or parameters of the model are altered.

Example: Dice Rolls

Example: Simulating Two Rolls of a Fair Four-Sided Die

We're going to roll two four-sided dice. Let X be the sum of two rolls, and Y be the larger of the two rolls. How would we simulate X and Y separately?

- Use the steps to run simulation for Y now

1. Set up

Define the probability space and related random variables and events, including assumptions.

- **Random variable.** $Y(\omega)$ is the larger of the two rolls in outcome ω max
- **Goal:** simulate Y , the larger of two rolls of a fair four-sided die
- **Sample space:** all possible outcomes of rolling two four-sided dice
 - Not necessary, but helpful to define the sample space

$$S = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 1), (2, 2), (2, 3), (2, 4), (3, 1), (3, 2), (3, 3), (3, 4), (4, 1), (4, 2), (4, 3), (4, 4)\}$$

- **Assumptions:** each die is fair and rolls are independent w/ replacement
 - Each outcome in S is equally likely with probability $1/16$

$$\begin{aligned} A &= (1, 1) \\ \hookrightarrow P(A) &= \frac{|A|}{|S|} = \frac{1}{16} \end{aligned}$$

2. Simulate

Run the simulation to generate outcomes according to the assumptions.

- Same code as for X , but we take the `max()` of each simulation instead of the `sum()`
- We also save the number of dice as `n_dice`, so it is easy to change later

```
1 reps <- 10000
2 n_dice <- 2
3
4 rolls <- expand_grid(sim = 1:reps, die = 1:n_dice) |>
5   mutate(roll = sample(1:4, n(), replace = TRUE))
6
7 Y_sims <- rolls |>
8   group_by(sim) |>
9   summarise(Y = max(roll))
```

Handwritten notes:
Line 3: `1-10000`
Line 4: `1-2`
Line 5: `sample(1:4, n(), replace = TRUE)` is underlined

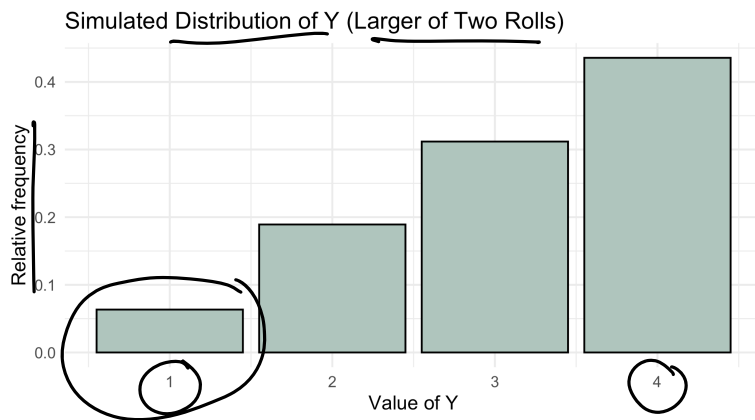
- We can look at the first 30 simulations

```
1 head(Y_sims$Y, 30)
[1] 4 2 4 3 2 2 3 4 4 4 2 4 4 4 2 3 4 4 1 4 4 3 4 4 2 4 4 4 1 4
```

3. Summarize

Analyze the output using plots and summary statistics like relative frequencies and averages.

► Show/Hide Code for plotting Y



If the problem asked us for something else, we could compute it:

- Average of Y
- Probability that $Y = 1$ $P(Y=1)$
- Probability that $Y > 3$ $P(Y>3)$

```
1 Y_sims |>
2   summarise(
3     average = mean(Y),
4     prob_1 = mean(Y == 1),
5     prob_over_3 = mean(Y > 3)
6   )
```

```
# A tibble: 1 × 3
  average prob_1 prob_over_3
  <dbl>   <dbl>   <dbl>
1   3.12  0.0634    0.436
```

4. Sensitivity analysis

Investigate how results change when assumptions or parameters of the model are altered.

What if we rolled three dice instead of two?

- We only need to change `n_dice`! Everything else stays the same (the changed line is highlighted)

```
1 reps <- 10000
2 n_dice <- 3
3
4 rolls <- expand_grid(sim = 1:reps, die = 1:n_dice) |>
5   mutate(roll = sample(1:4, n(), replace = TRUE))
6
7 Y_sims <- rolls |>
8   group_by(sim) |>
9   summarise(Y = max(roll))
```

①

②

③

4. Sensitivity analysis

Investigate how results change when assumptions or parameters of the model are altered.

- What changed? Each simulation now has **three rolls**, like the first one here: ► Show/Hide Code for plotting Y

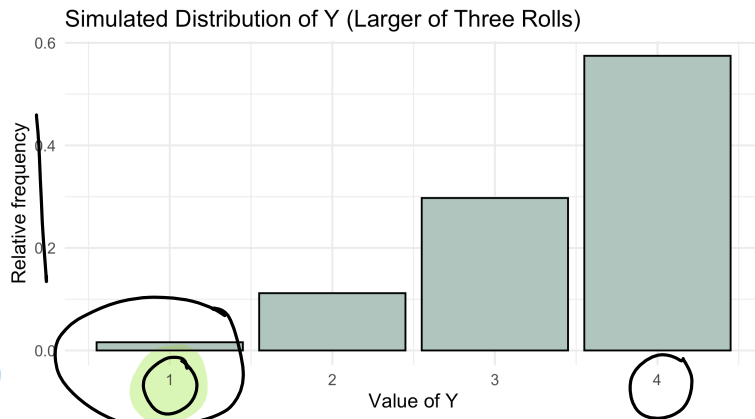
```
1 rolls |> filter(sim == 1) ①
```

A tibble: 3 × 3

	sim	die	roll
	<int>	<int>	<int>
1	1	1	2
2	1	2	4
3	1	3	2

- With more dice, it is more likely that **at least one** die shows a 4

- $P(Y = 4)$ goes from about 0.44 with two dice to about 0.58 with three dice



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